Exploratory Data Analysis (EDA)

Data set: House prices in New York
Dependent variable: House price
Predictors: Age, Land.Value, Living.Area, Central.Air

Checking Missingness of Data

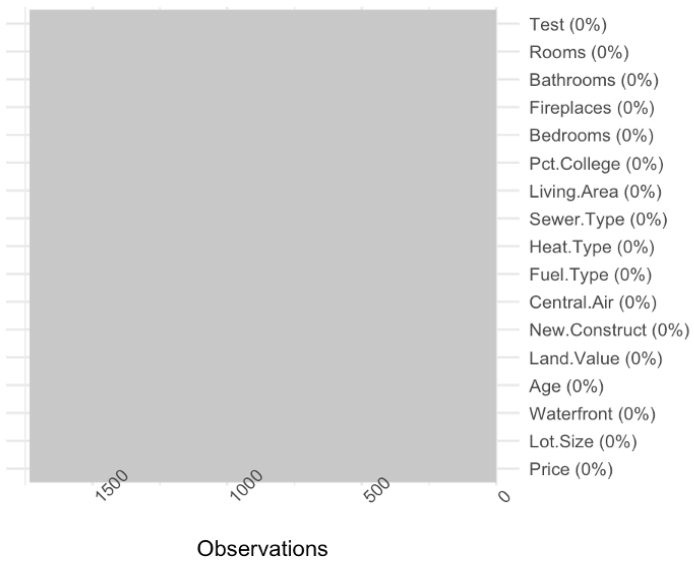

Figure 1: Missing data pattern in data set. No missing observations were found.

Descriptive Statistics

	0 - No Central Air (N=1099)	1 - Central Air (N=635)
Price		
Mean (SD)	186000 (78400)	255000 (114000)
Median [Min, Max]	171000 [10300, 670000]	235000 [5000, 775000]
Age		
Mean (SD)	34.7 (34.3)	17.1 (14.2)
Median [Min, Max]	22.0 [0, 225]	16.0 [0, 124]
Land.Value		
Mean (SD)	28200 (27800)	45600 (42500)
Median [Min, Max]	20300 [300, 250000]	34500 [200, 413000]
Living.Area		
Mean (SD)	1620 (565)	1990 (640)
Median [Min, Max]	1510 [616, 4490]	1890 [840, 5230]

Table 1: Descriptive statistics of Price, Age, Land.Value, Living.Area by Central.Air

Initial Visualisations

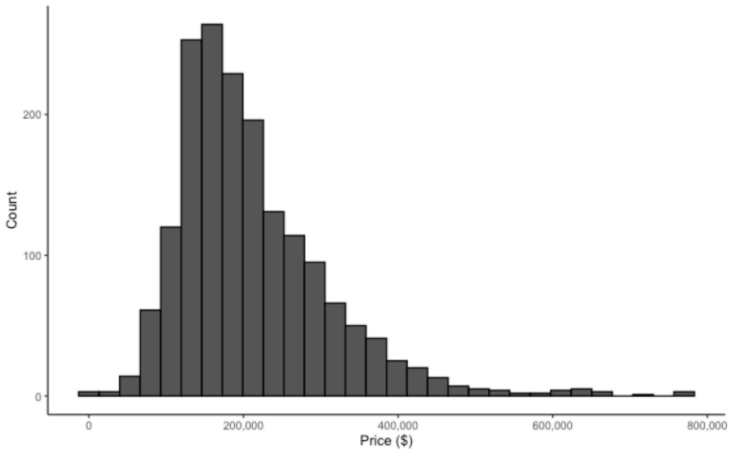

Figure 2: Histogram of the distribution of House Prices in New York

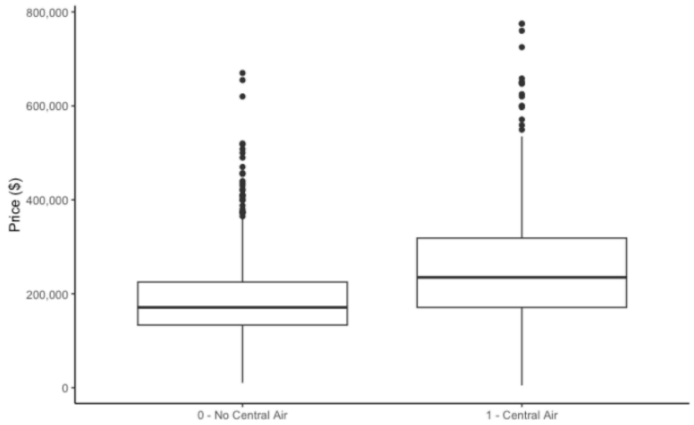

Figure 3: Box plot of the distribution of Prices by Central Air

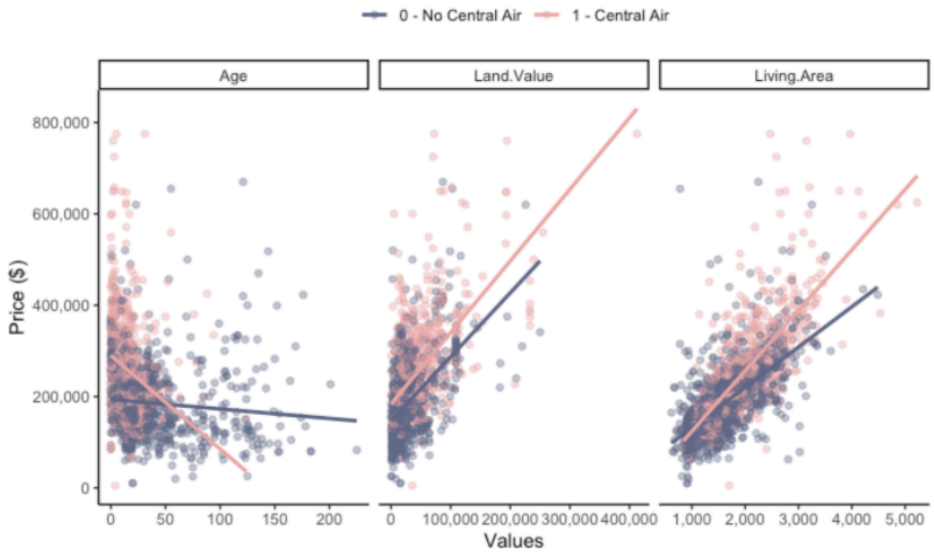

Figure 4: Faceted scatter plots for Prices against Age, Land.Value, Living.Area grouped by Central.Air

Summary

From my initial EDA it looks like Central.Air is an important factor for predicting the house price. As shown in Figure 3, houses with central air generally have higher price distributions than houses with no central air. Age, Land.Value and Living.Area appear to also have a relationship with the house prices. In Figure 4, we can observe that older houses tend to have lower prices, while houses with larger Land.Value and Living.Area tend to be more expensive.